Integer Linear Programming

Further Problems

Sensitivity: Internal

Minimum Set Cover

Sensitivity: Internal

General Set Covering Problem

minimise

$$\sum_{t \in S} x_t$$

subject to:

$$\sum_{t:e \in t} x_t \geq 1, \qquad \forall e \in U$$

$$x_t \in \{0,1\}$$

S – All possible sets to choose from

$t \in S$ – A given set t

x_t - Set t chosen or not

N – number of sets |S|

U = Set of all elements contained in S

Sensitivity: Internal

General Set Covering Problem

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$$\sum_{t \in S} x_t$$

subject to:

$$\sum_{t:e \in t} x_t \geq 1\,, \qquad \forall e \in U$$

$$x_t \in \{0,1\}$$

Example

$$a = \{1,2,5\}, \qquad b = \{2,3\},$$
$$c = \{4,5\}, \qquad d = \{3,4\}$$

$$S = \{a, b, c, d\}$$
$$U = \{1,2,3,4,5\}$$
$$N = 4$$

Pick the minimum number of sets to get all elements in U

Answer by inspection?

Sensitivity: Internal

General Set Covering Problem

minimise

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subject to:

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$$x_t \in \{0,1\}$$

$a = \{1,2,5\}, \quad b = \{2,3\}, \quad c = \{4,5\}, \quad d = \{3,4\}$

$S = \{a, b, c, d\}$
$U = \{1,2,3,4,5\}$
$N = 4$

minimise

$x_a + x_b + x_c + x_d$

subject to:

$x_a \geq 1$	(Element 1)
$x_a + x_b \geq 1$	(Element 2)
$x_b + x_d \geq 1$	(Element 3)
$x_c + x_d \geq 1$	(Element 4)
$x_a + x_c \geq 1$	(Element 5)

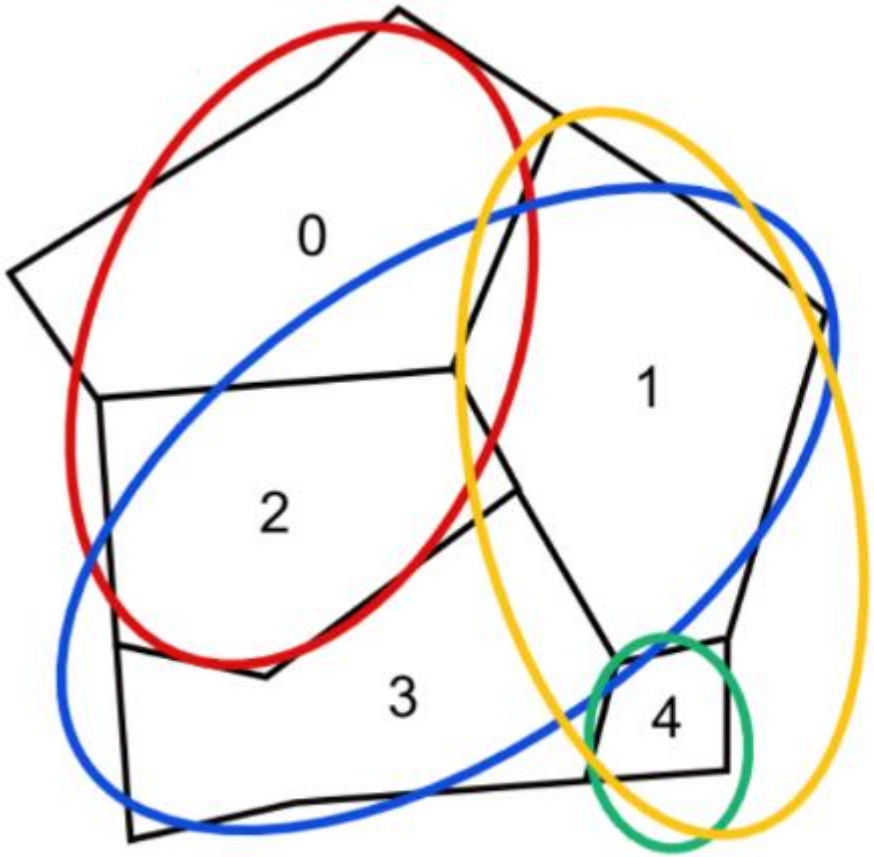
$x_t \in \{0,1\}$

Minimum set cover 2

$\{a, d\}$

Sensitivity: Internal

Costed Fire Station Example



S – All possible fire stations to choose from
$t \in S$ – A given fire station t
x_t - fire station t chosen or not
c_t - cost of running fire station t
$U =$ Set of all cities contained in S

$$S = \{a, b, c, d\}$$
$$U = \{0,1,2,3,4\}$$
$$N = 4$$

$$s_a = \{0,2\}$$
$$s_b = \{1,2,3\}$$
$$s_c = \{1,4\}$$
$$s_d = \{4\}$$

$$c_a = 12$$
$$c_b = 7$$
$$c_c = 10$$
$$c_d = 5$$

Sensitivity: Internal

Costed Fire Station Example

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$$c_a = 12$$
$$c_b = 7$$
$$c_c = 10$$
$$c_d = 5$$

minimise

$$\sum_{t \in S} c_t x_t$$

subject to:

$$\sum_{t:e \in t} x_t \geq 1, \qquad \forall e \in U$$

$$x_t \in \{0,1\}$$

minimise

$$c_a x_a + c_b x_b + c_c x_c + c_d x_d$$

subject to:

$$x_a \geq 1 \quad \text{(City 0 covered)}$$
$$x_b + x_c \geq 1 \quad \text{(City 1 covered)}$$
$$x_a + x_b \geq 1 \quad \text{(City 2 covered)}$$
$$x_b \geq 1 \quad \text{(City 3 covered)}$$
$$x_c + x_d \geq 1 \quad \text{(City 4 covered)}$$
$$x_t \in \{0,1\}$$

Sensitivity: Internal

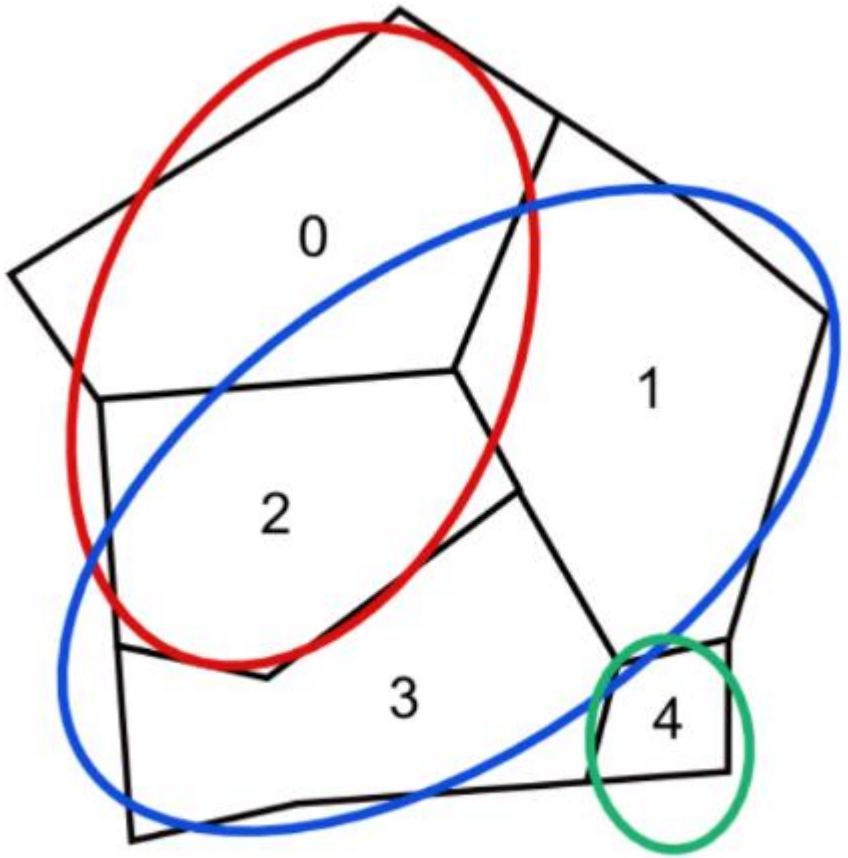
Costed Fire Station Example

S – All possible fire stations to choose from
$t \in S$ – A given fire station t
x_t - fire station t chosen or not
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U = Set of all cities contained in S

$$S = \{a, b, c, d\}$$
$$U = \{0,1,2,3,4\}$$

$$s_a = \{0,2\}$$
$$s_b = \{1,2,3\}$$
$$s_c = \{1,4\}$$
$$s_d = \{4\}$$

$$c_a = 12$$
$$c_b = 7$$
$$c_c = 10$$
$$c_d = 5$$



cost = 12 + 7 + 5 = 24

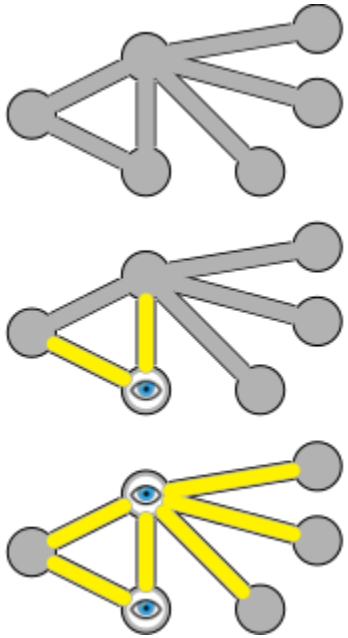
Sensitivity: Internal

Minimum Vertex Cover

Sensitivity: Internal

Vertex Cover

A set of vertices that includes at least one endpoint of every edge



Sensitivity: Internal

Minimum Vertex Cover



Node variables x_u

Edge variables x_{uv}

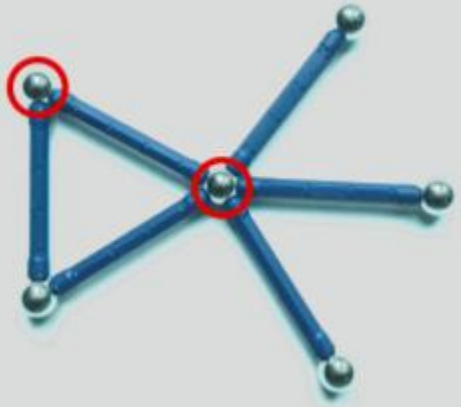
Labels ℓ_v

Example 2: Min. vertex cover

$$\min \sum_{u \in V} x_u$$

s.t.

$$\forall \{u, v\} \in E: x_u + x_v \geq 1$$



Sensitivity: Internal

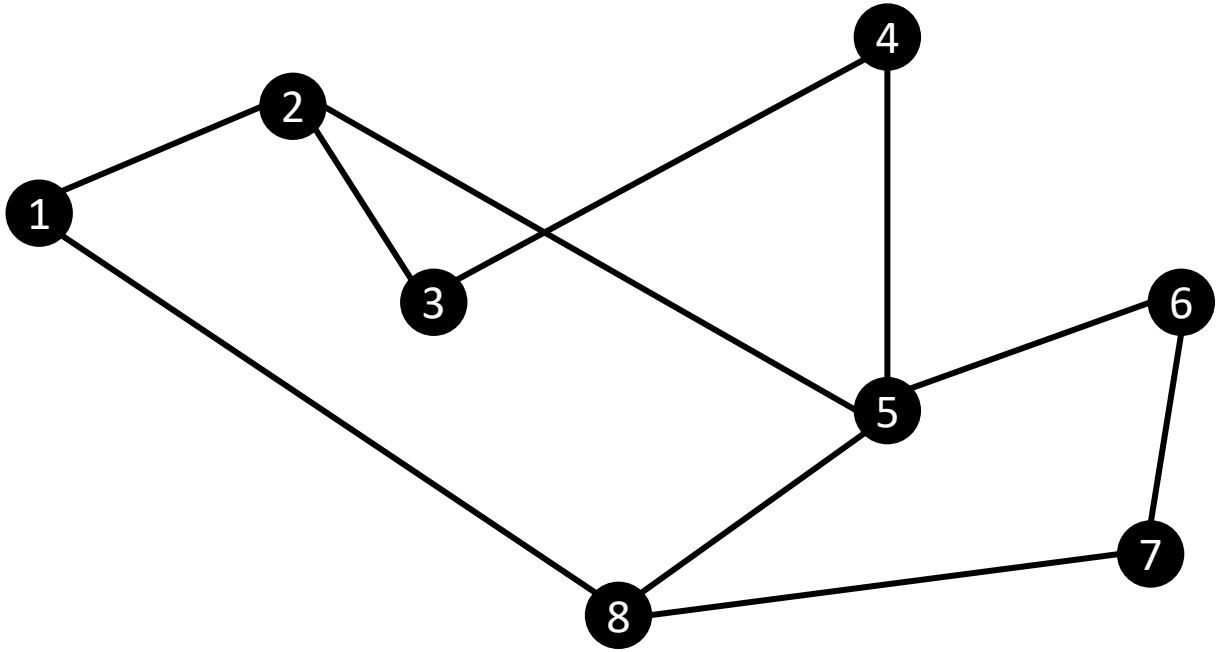
Example – Installing Security Cameras

- You wish to install security cameras in a school to cover every corridor.
- Obviously you would like to install as few as needed

- The following example is adapted from https://brilliant.org/wiki/vertex-cover/

Sensitivity: Internal

Example – Installing Security Cameras



minimise

$$\sum_{u \in V} x_u = x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 + x_8$$

subject to:

$$x_1 + x_2 \geq 1$$
$$x_1 + x_8 \geq 1$$
$$x_2 + x_3 \geq 1$$
$$x_2 + x_5 \geq 1$$
$$x_3 + x_4 \geq 1$$
$$x_4 + x_5 \geq 1$$
$$x_5 + x_6 \geq 1$$
$$x_5 + x_8 \geq 1$$
$$x_6 + x_7 \geq 1$$
$$x_7 + x_8 \geq 1$$

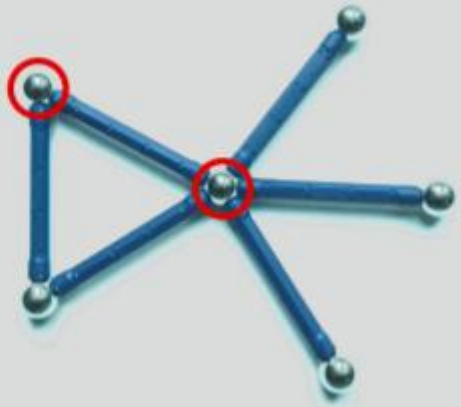
$$x_u \in \{0,1\}$$

Example 2: Min. vertex cover

$$\min \sum_{u \in V} x_u$$

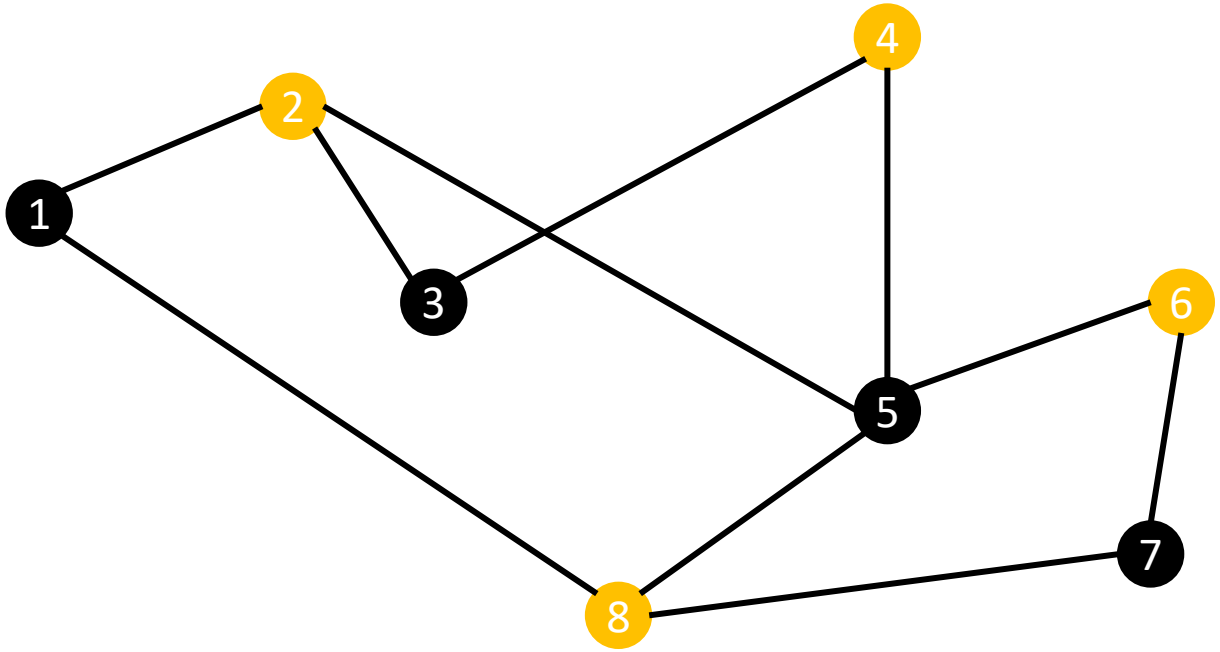
s.t.

$$\forall \{u, v\} \in E: x_u + x_v \geq 1$$



Sensitivity: Internal

Example – Installing Security Cameras



minimise

$$\sum_{u \in V} x_u = x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 + x_8$$

subject to:

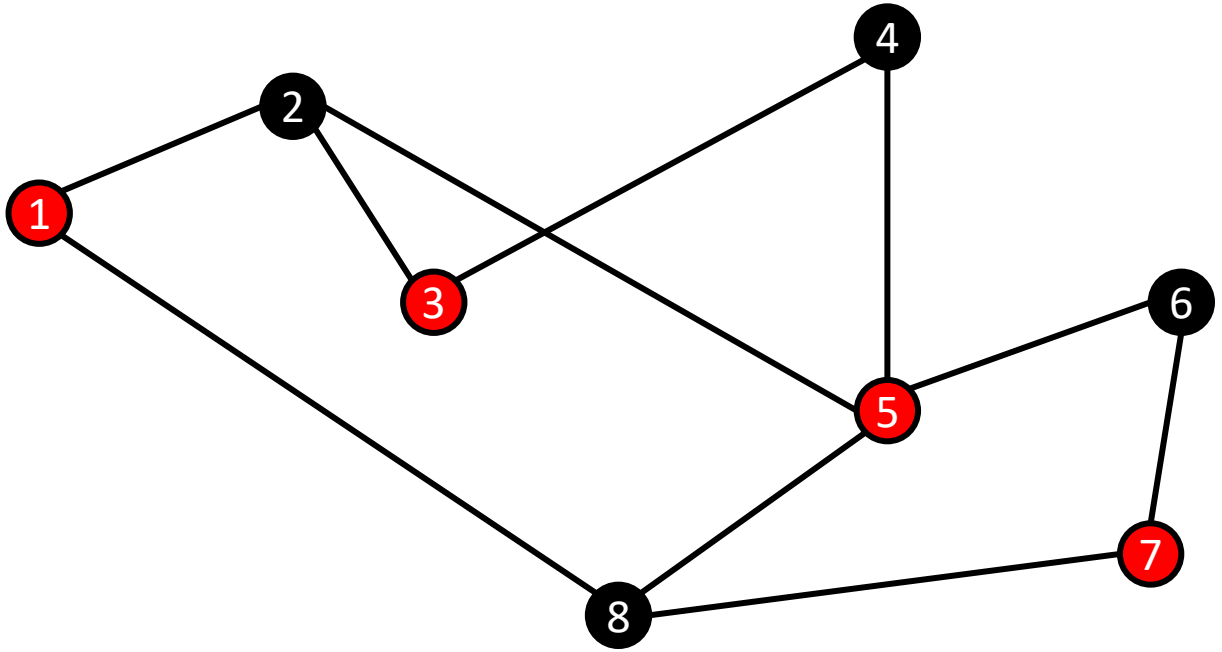
$$x_1 + x_2 \geq 1$$
$$x_1 + x_8 \geq 1$$
$$x_2 + x_3 \geq 1$$
$$x_2 + x_5 \geq 1$$
$$x_3 + x_4 \geq 1$$
$$x_4 + x_5 \geq 1$$
$$x_5 + x_6 \geq 1$$
$$x_5 + x_8 \geq 1$$
$$x_6 + x_7 \geq 1$$
$$x_7 + x_8 \geq 1$$

$$x_u \in \{0,1\}$$

MATLAB Solution

Sensitivity: Internal

Example – Installing Security Cameras



Another possibility

minimise

$$\sum_{u \in V} x_u = x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 + x_8$$

subject to:

$$x_1 + x_2 \geq 1$$
$$x_1 + x_8 \geq 1$$
$$x_2 + x_3 \geq 1$$
$$x_2 + x_5 \geq 1$$
$$x_3 + x_4 \geq 1$$
$$x_4 + x_5 \geq 1$$
$$x_5 + x_6 \geq 1$$
$$x_5 + x_8 \geq 1$$
$$x_6 + x_7 \geq 1$$
$$x_7 + x_8 \geq 1$$

$$x_u \in \{0,1\}$$

Sensitivity: Internal

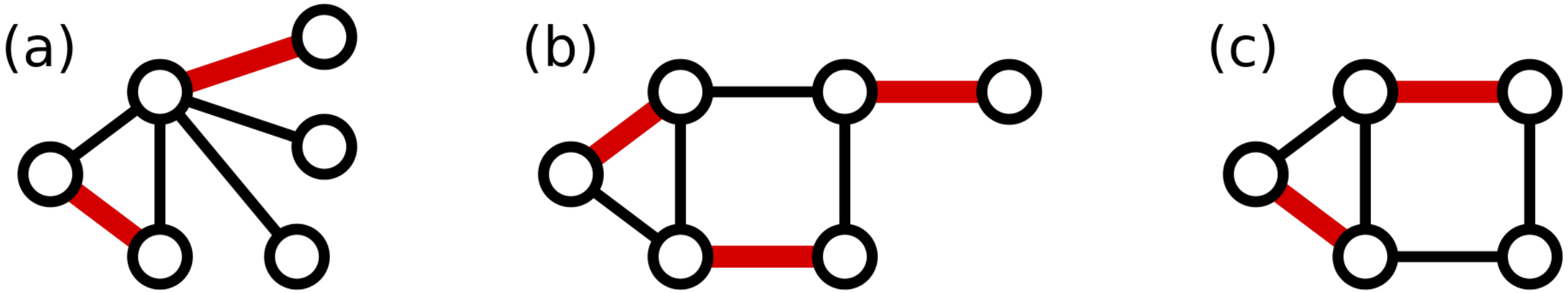
Maximum Weight Matching and Assignment Problem

Sensitivity: Internal

Maximum Matching

- Matching – Set of edges without common vertices
- Maximum Matching – Matching with the largest number of edges

Examples of Maximum Matchings

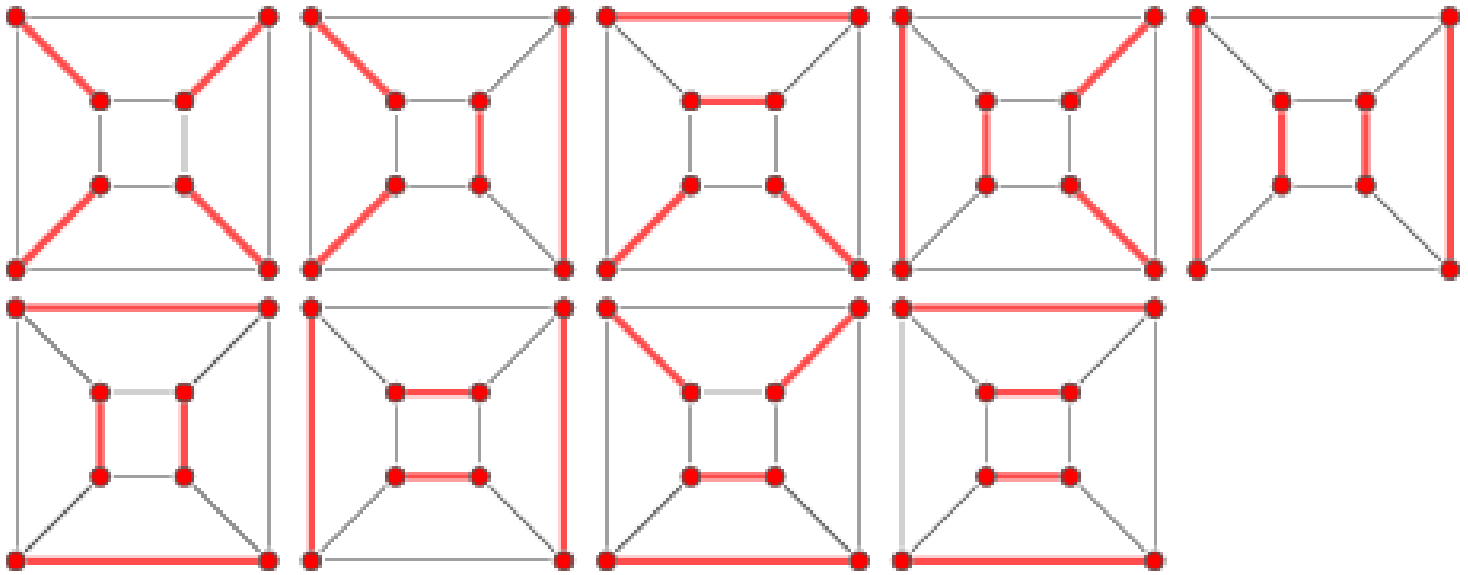


Sensitivity: Internal

Perfect Matching

- Matching – Set of edges without common vertices
- Perfect Matching – Matching that includes every vertex

Examples



Sensitivity: Internal

Maximum Weight Matching



Node variables x_u

Edge variables x_{uv}

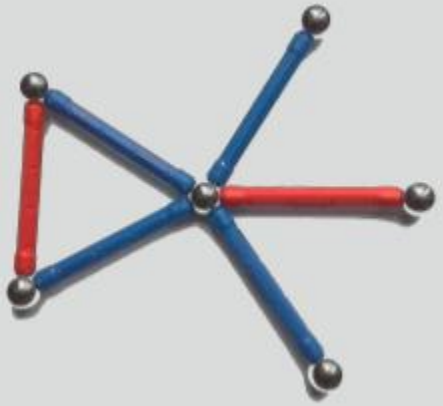
Labels ℓ_v

Example 1: Max. weight matching

$$\max \sum_{\{u,v\} \in E} w_{uv} \cdot x_{uv}$$

s.t.

$$\forall u \in V: \sum_{\{u,v\} \in E} x_{uv} \leq 1$$



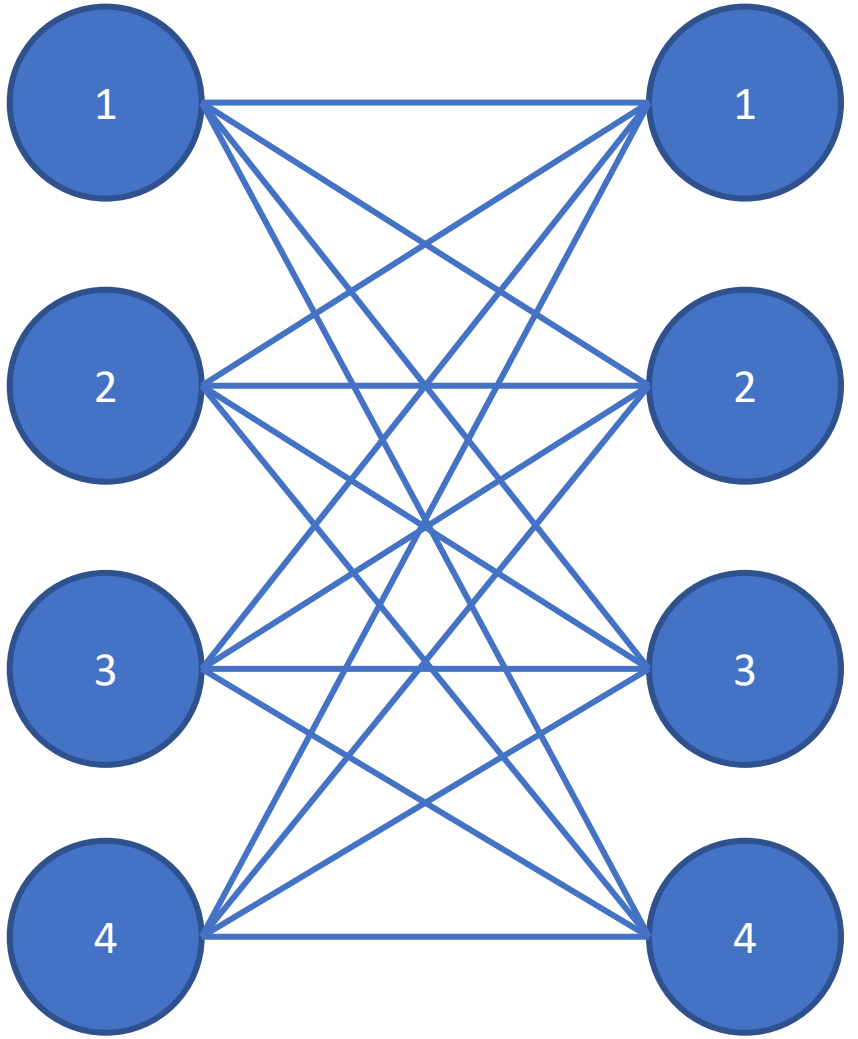
- Select a set of edges without common vertices
- Maximise the weight of the edges

Sensitivity: Internal

Balanced Assignment Problem

- Assign a worker to each job – minimise completion time

	Job 1	Job 2	Job 3	Job 4
Worker 1	9	2	7	8
Worker 2	6	4	3	7
Worker 3	5	8	1	8
Worker 4	7	6	9	4



This is a bipartite graph and requires a perfect matching…

How many solutions??

$4!$

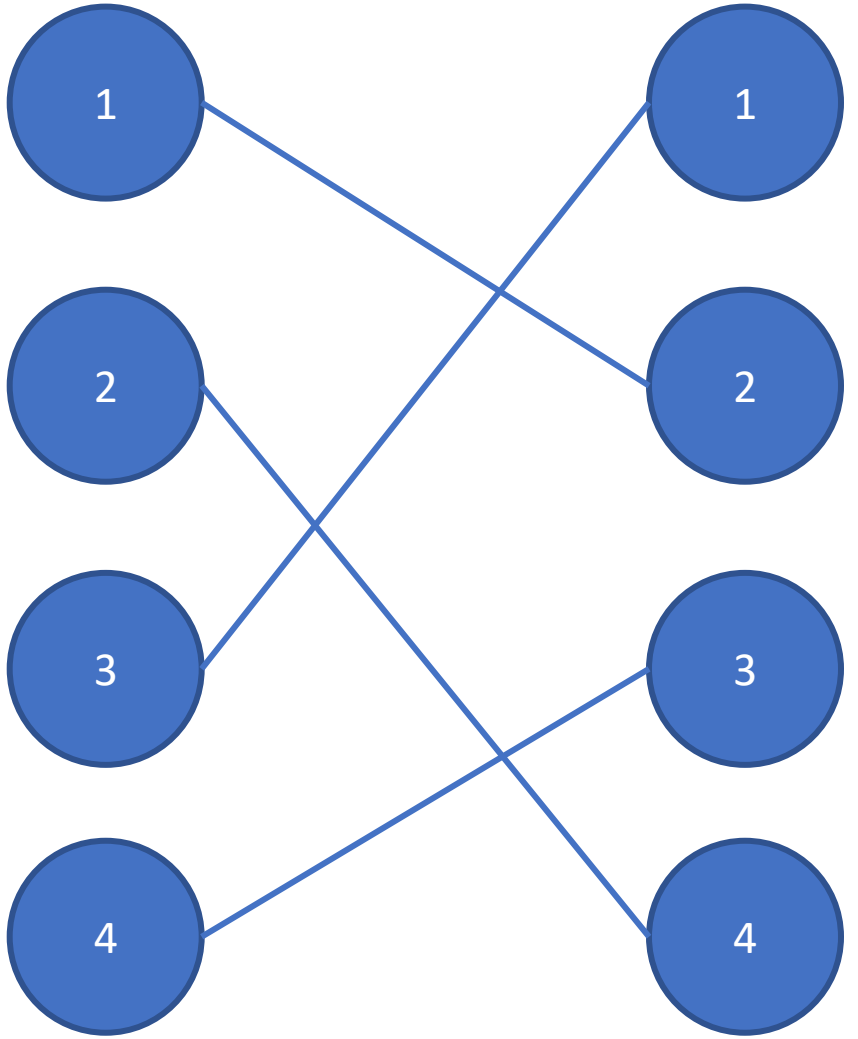
In general $n!$

Sensitivity: Internal

Balanced Assignment Problem

- Example Solution

	Job 1	Job 2	Job 3	Job 4
Worker 1	9	2	7	8
Worker 2	6	4	3	7
Worker 3	5	8	1	8
Worker 4	7	6	9	4



This is a bipartite graph and requires a perfect matching…

Here every vertex is included and no edge has a common vertex

Cost is $2 + 7 + 5 + 9 = 23$

Sensitivity: Internal

Balanced Assignment Problem – Model

- Pick 1 per row and 1 per column

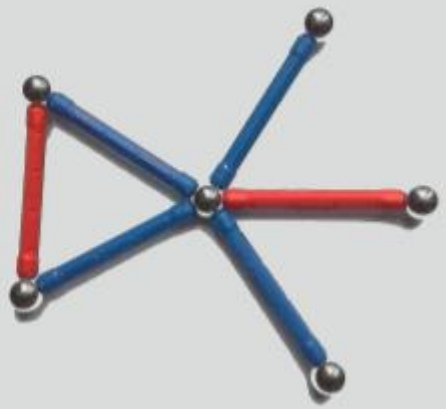
	Job 1	Job 2	Job 3	Job 4
Worker 1	9	2	7	8
Worker 2	6	4	3	7
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Worker 4	7	6	9	4

Example 1: Max. weight matching

$$\max \sum_{\{u,v\} \in E} w_{uv} \cdot x_{uv}$$

s.t.

$$\forall u \in V: \sum_{\{u,v\} \in E} x_{uv} \leq 1$$



W - Set of Workers
J - Set of Jobs
x_{uv} - Worker u completes job v
c_{uv} - Completion time for worker u completing job v

minimise

$$\sum_{u \in W} \sum_{v \in J} c_{uv} x_{uv} = c_{11}x_{11} + c_{12}x_{12} + \cdots + c_{uv}x_{uv} + \cdots + c_{44}x_{44}$$

subject to:

$$\sum_{v \in J} x_{uv} = 1, \qquad \forall u \in W$$

$$\sum_{u \in W} x_{uv} = 1, \qquad \forall v \in J$$

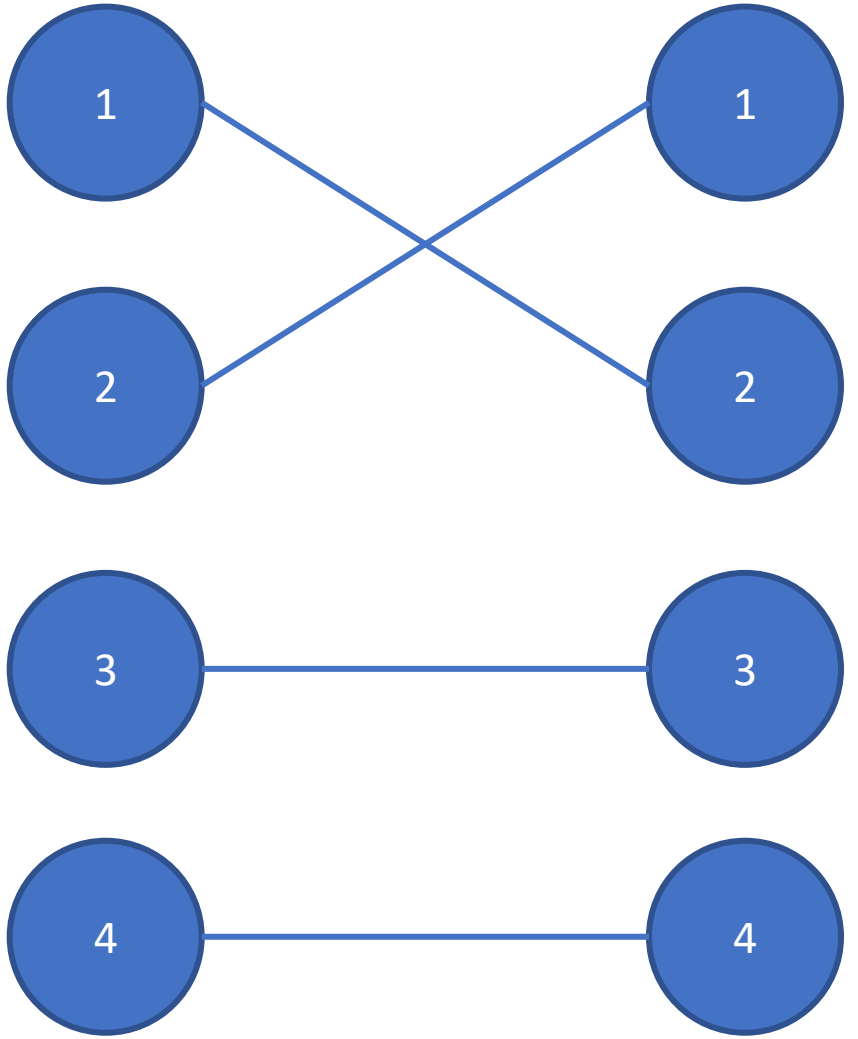
$$x_{uv} \in \{0,1\}$$

Sensitivity: Internal

Balanced Assignment Problem

- Optimal Solution

	Job 1	Job 2	Job 3	Job 4
Worker 1	9	2	7	8
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Worker 3	5	8	1	8
Worker 4	7	6	9	4



This is a bipartite graph and requires a perfect matching…

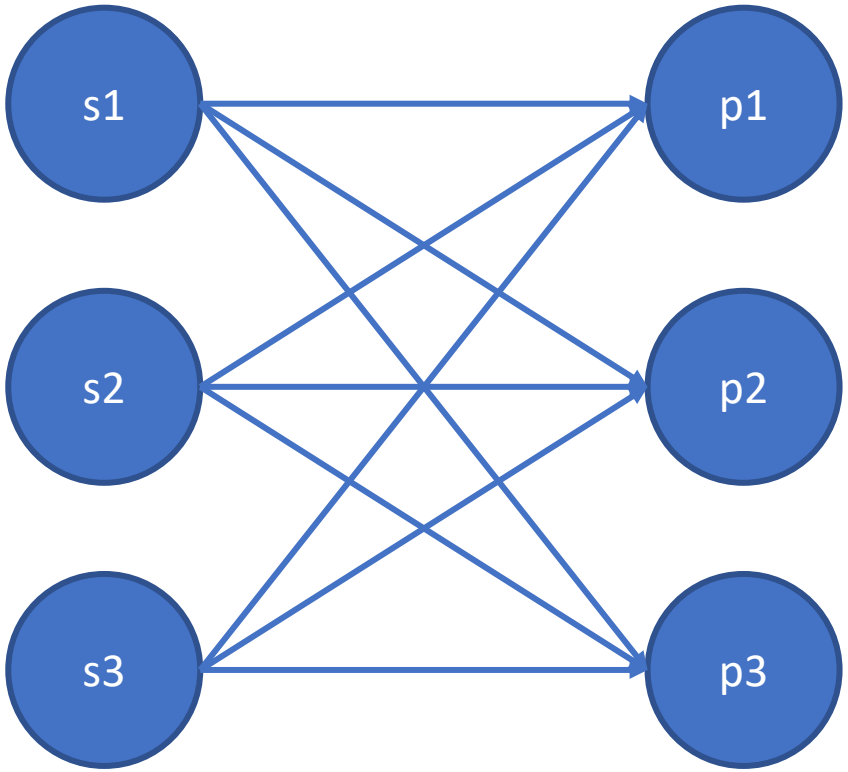
Here every vertex is included and no edge has a common vertex

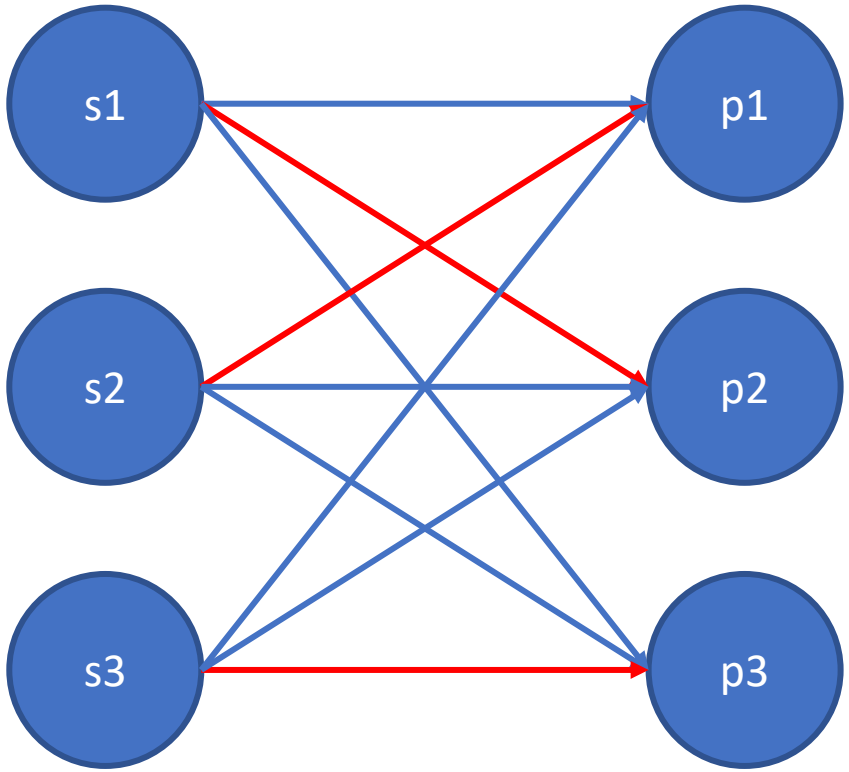
Cost is $2 + 6 + 1 + 4 = 13$

Sensitivity: Internal

Assignment Problem

Assign 3 students to the 3 different dissertation projects.





Sensitivity: Internal

Tutorial

1. Study these models and understand the Python code. It will be useful for coming lectures.
2. Code the costed fire station in Pyomo using an incidence matrix (matrix of 0s and 1s). Think about how each element (city) is covered by each fire station.

3. Look at the examples on Gurobi - https://www.gurobi.com/resource/modeling-examples-using-the-gurobi-python-api-in-jupyter-notebook/

4. Pick an example and try coding it in Pyomo

5. Try coding the problems below yourself.

Look up other problems, such as:

- Maximum Set Cover
- Minimum Edge Cover
- Maximum Independent Set

Sensitivity: Internal